

Testing Models Relating Rejection, Depression, Interpersonal Needs, and Psychache to Suicide Risk in Nonclinical Individuals

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Objectives: Using structural equation modeling, we tested a primary model of suicide risk and 3 competing, alternative models based on 4 psychological variables deemed important in the literature (perception of parental rejection, depression, interpersonal needs comprising perceived burdensomeness and thwarted belongingness, and psychache), in a nonclinical sample of Portuguese adults.

Method: A convenience sample of 203 adults (100 men, 103 women; aged 18–65 years) participated in this study. **Results:** Analyses demonstrated that the proposed primary model had the best fit to the observed data. The differences in fit indexes for this model and one of the alternative models, however, were not substantial. **Conclusion:** Perceived parental rejection related directly to suicide risk and indirectly via depression and interpersonal needs. Depression linked indirectly to suicide risk via interpersonal needs and psychache. Interpersonal needs related directly to suicide risk and indirectly via psychache, which related directly to suicide risk. © 2015 Wiley Periodicals, Inc. *J. Clin. Psychol.* 71:994–1003, 2015.

Keywords: rejection; depression; psychache; interpersonal needs; suicide risk; community-based sample

Suicide is a major health problem across the lifespan for both clinical and nonclinical populations. Although suicide is a low base-rate phenomenon, it is an important mental health issue (World Health Organization, 2012), and, as such, the identification of relevant factors associated with suicide risk is a critical research endeavour. In Portugal, over 1,000 people die by suicide every year (10.3 per 100,000 in the population; National Institute of Statistics [NIS], 2013) and suicide attempts are estimated to have a rate 10 to 20 times higher than deaths by suicide (Fotti, Katz, Afifi, & Cox, 2006).

Furthermore, suicidality is not only an issue for Portugal but also a significant public health concern worldwide. In Portugal, suicide is a demographic-based phenomenon, with higher rates in the southern part of the country, the Alentejo region (NIS, 2013). It is not completely clear why this regional difference (endemic phenomenon in Alentejo) occurs, but a lower level of religiosity compared with the northern part of the country may be one possible explanation. Although it is recognized that suicide takes place within a specific cultural context, in the present study, we focus on psychological variables, rather than sociocultural factors. As such, our study may be of an etic nature.

The prediction of suicide remains extremely difficult (Overholser, Braden, & Dieter, 2012) because suicide behaviors are regarded as multifactorial phenomena (e.g., Hawton & van Heeringe, 2009). Psychological perspectives of suicide (e.g., Goldney & Schioldann, 2004) have demonstrated the existence of potential interactions among various individual psychological risk factors (e.g., Beautrais, Collings, Ehrhardt, & Henare, 2005). In particular, several psychological variables that have been empirically related to suicide risk include the perception of parental rejection (e.g., Campos, Besser, & Blatt, 2013), Shneidman's (1993) concept of psychache (e.g., Troister

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& Holden, 2012), and Joiner's (2005) interpersonal needs constructs. Depression has also been related to suicide risk (e.g., Moscicki, 1999; Youssef et al., 2004). Although a few studies have attempted to demonstrate how these four factors combined relate to indicate suicide risk (e.g., Campos et al., 2013; Campos & Holden, in press), none of them have simultaneously tested all four of these variables.

This study integrates into one model the contributions of three different perspectives on suicide, a more focused psychodynamic approach, and Joiner's (2005) and Shneidman's models (1979, 1993). We also incorporate the contribution of depression because empirical research has systematically demonstrated its importance for suicide. Suicidality, in fact, is one of the diagnostic criteria for a depressive disorder. Depression may also be a mediator between distal and trait risk variables and suicide risk. It is important to realize, however, that not all depressed people will present suicidal ideation or will try to die by suicide.

Several psychodynamic theorists and researchers (e.g., Blatt, 2004) advocate that when disruptive relationships between children and caregivers occur, the consequent internalized dysfunctional internal representations result in a vulnerability to future psychopathology, especially depression (e.g., Blatt, 2004; Blatt & Homann, 1992; Campos et al., 2013; Titelman, Nilsson, Svensson, Karlsson, & Bruchfeld, 2011). Empirical research has found a relationship between dysfunctional parental representations and insecure attachment and suicidality (Barksdale, Walrath, Compton, & Goldston, 2009; Gutierrez, Osman, Kooper, Barrios, & Bagge, 2000; Palitsky, Mota, Afifi, Downs, & Sareen, 2013; Titelman et al., 2011). According to Heo (2008), depression mediated the relationship between parental representations and suicidal ideation. Titelman et al. (2011) found that, when comparing individuals having a history of suicidal attempts with a control group, the absence of a secure attachment discriminated between groups. Campos et al. (2013) verified that recollections of parental rejection were significantly associated with suicidality and indirectly related to suicidality via depressive symptoms.

Recently, the theories of Joiner (Joiner, Van Orden, Witte, & Rudd, 2009; Van Orden et al., 2010) and Shneidman (1993) have garnered substantial empirical support for explaining suicide behaviors. Psychache is an affective state of anguish and despair that is distinct from the cognitive constructs of depression and hopelessness (DeLisle & Holden, 2009). Shneidman (1993) has suggested that extreme mental pain (i.e., psychache) is responsible for suicide behaviors and that all other variables relate to suicide only via psychache. Associations between psychache and suicidality (Holden et al., 2001; Troister & Holden, 2010, 2013) and support for psychache as being essential for suicide risk have been established with empirical research showing psychache's full mediation effects, for example, between alexithymia and suicidality (Keefer, Holden, & Gillis, 2009).

The interpersonal-psychological theory of suicide (Joiner, 2005; Joiner et al., 2009) indicates that three psychological factors are required for suicide to occur: perceived burdensomeness, thwarted belongingness, and the acquired capability for suicide, all of which are also related to several psychological disorders, including depression (Silva, Ribeiro, & Joiner, 2015). Because we used the Interpersonal Needs Questionnaire in the present study, we did not evaluate acquired capability. Perceived burdensomeness is an important predictor of suicidality, mediating, for example, the relationship between depression and suicide ideation among older adults (Jahn, Cukrowicz, Linton, & Prabhu, 2011). Associations linking perceived burdensomeness and thwarted belongingness with suicidality have also been found among adult populations (Lester & Gunn, 2012; O'Keefe et al., 2014). Perceived burdensomeness and thwarted belongingness as well as psychache fully mediated the association between depression and suicide risk (Campos & Holden, in press).

Aim of the Study

The aim of the present study was to evaluate, using structural equation modeling, a model of suicide risk in a nonclinical sample of Portuguese adults, based on the following four psychological variables deemed important in the literature: perception of parental rejection, depression, interpersonal needs (perceived burdensomeness and thwarted belongingness), and psychache. This study brings together, in a unified model, three perspectives on suicide risk,

Model 1

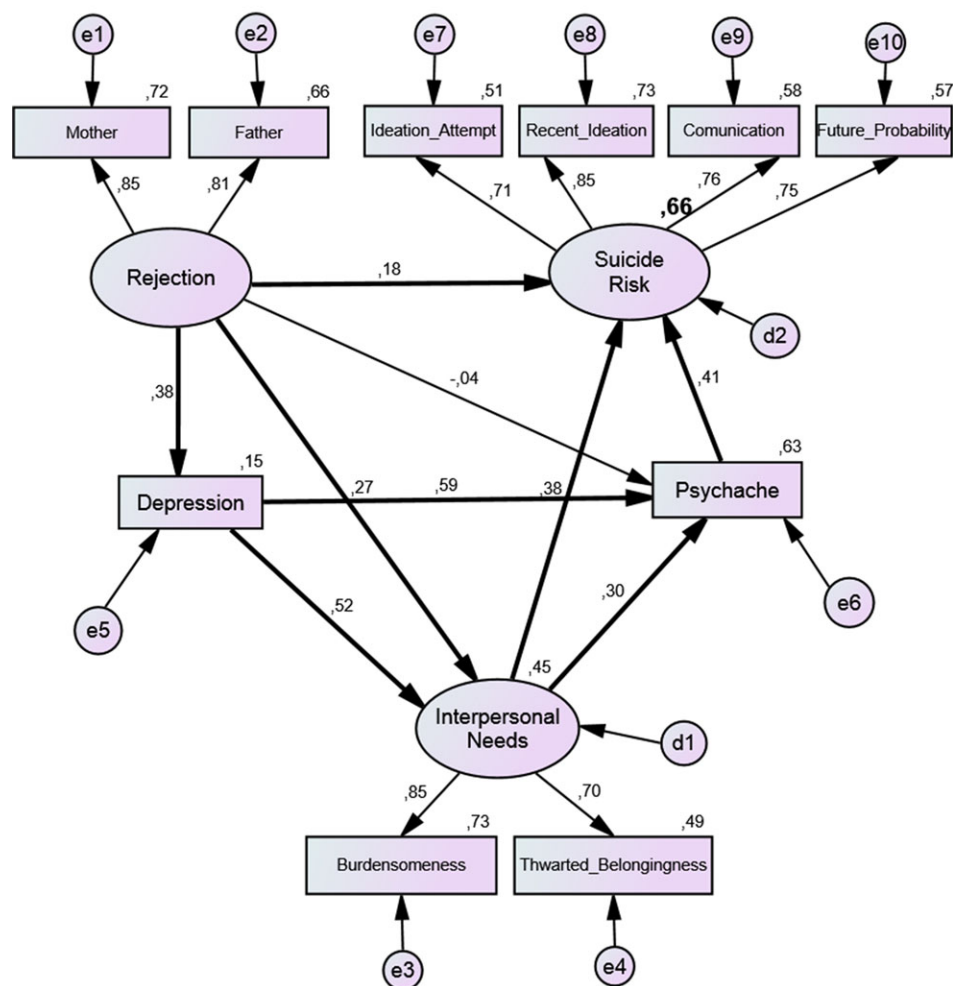


Figure 1. Tested model.

Note. This model conceptualizes direct and indirect effects (unidirectional arrows) of the studied variables for predicting suicide risk. Rectangles are observed variables. Rejection, interpersonal needs, and suicide risk are latent variables (circles) that are evaluated with two, two, and four indicators, respectively. The dark bold arrows represent statistically significant relationships between variables. The values next to the variables indicate the proportion of explained variance (R^2).

a more psychodynamic-focused tradition and Shneidman and Joiner's perspectives. Given the exploratory nature of the present study, three alternative competing models were also tested. The following four clinically important indicators measured suicide risk: ideation and attempt, recent ideation, communication of intention, and estimate of future probability. A history of suicidality, including ideation, reported intention, and past attempts are considered major factors of suicide risk (e.g., Ash, 2006; Bolognini, Plancherel, Laget, & Halfon, 2003; Joiner et al., 2005; King et al., 2001).

In our proposed primary model (see Figure 1), perceived parental rejection relates directly to suicide risk and indirectly via depression, interpersonal needs, and psychache. In this model, depression relates only indirectly to suicide risk via interpersonal needs and psychache. Interpersonal needs links directly to suicide risk and indirectly via psychache, which relates directly to

suicide. This model is derived from the literature, in which rejection relates directly to suicide risk and indirectly via depression (Campos et al., 2013). However, it is logical to expect that rejection would also relate to interpersonal needs and psychache. Feeling rejected by one's parents should be related to feeling isolated, like being a burden to others, and psychological pain. Furthermore, based on the literature, depression should also relate directly to interpersonal needs and psychache and indirectly to suicide risk via those variables (Campos & Holden, in press). It is also logical that interpersonal needs would relate to psychache. Feeling like being a burden to others and not belonging with others would inevitably relate to psychological pain.

In addition to this primary model, we tested three alternative competing models that were theoretically and/or empirically logical. In Model 2, among the four constructs of parental rejection, depression, interpersonal needs, and psychache, only psychache was postulated to directly link to suicide risk. This expectation was based on Shneidman's model for the exclusivity of psychache as a cause of suicide. Model 3 is similar to our primary model but does not propose a relationship between rejection and psychache or between interpersonal needs and psychache because these relationships have not been empirically demonstrated previously.

Finally, Model 4 is similar to Model 3 except interpersonal needs is treated as two independent indicators, with only perceived burdensomeness relating directly to suicide risk and thwarted belongingness contributing indirectly via perceived burdensomeness. This is based on recent empirical efforts suggesting that the relationships between thwarted belongingness and suicidality is weaker than the relationships between perceived burdensomeness and suicidality (e.g., O'Keefe et al., 2014) and that thwarted belongingness only affects suicidality via perceived burdensomeness (e.g., Wong, Koo, Tran, Chiu, & Mok, 2011).

Methods

Participants and Procedures

A convenience sample of 203 adults (100 men, 103 women) living in various Portuguese regions participated in this study. Participants were aged between 18 and 65 years (mean [M] = 37.86, standard deviation [SD] = 11.68), and their education ranged from 6 to 21 years of schooling (M = 11.68, SD = 3.16). A minority of participants (26.1%) was unemployed and 53.2% were married or living with a romantic partner.

Participants responded to a request for volunteers to take part in a research study concerning suicide risk. Trained research assistants contacted the respondents in public settings such as coffee shops, other small shops, and churches. Of the 254 individuals initially contacted, 37 declined to participate. Because of constraints related to research assistant availability, data collection was discontinued after the 254 contacts. A further 14 protocols of individuals initially interviewed were eliminated due to missing sociodemographic information, an elevated number of missing responses, or not reporting information regarding their father or mother because the participant never knew that parent. The final sample thus comprised 203 volunteers who participated after signing an informed consent form. Volunteers were not compensated for their participation.

All protocols contained a sociodemographic questionnaire and the five measures that assessed rejection, depression, interpersonal needs, psychache, and suicide risk. Trained research assistants collected protocols individually and instructions were provided in written form. Participants were given the opportunity to discontinue their participation at any time. The questionnaires were presented in a counterbalanced order but the sociodemographic questionnaire was always first. Data collection fulfilled all recommended ethical guidelines of the Portuguese Psychologists Board and the American Psychological Association.

Measures

Inventory for Assessing Memories of Parental Rearing Behavior (EMBU; Perris, Jacobsson, Linndström, von Knorringand, & Perris, 1980). The EMBU assesses adults' memories of their parents' child-rearing behaviors. The questionnaire yields six scales: Rejection,

Emotional Support and Overprotection for each of the father and mother. In the present study, the EMBU was used to assess perceived parental (mother and father) rejection, which relates to the attitudes of the parents that aim to change the child's wishes and that are felt as rejection.

In the Portuguese adaptation (Canavarro, 1996), the scale scores are based on the six factors and present acceptable levels of internal consistency and test-retest reliability (correlations of .72 and .83 for the rejection scales over a 6-week interval). In the present study, Cronbach's alpha reliabilities for Father Rejection and Mother Rejection scale scores were .82 and .85, respectively.

The Center for the Epidemiological Studies of Depression Scale (CES-D; Radloff, 1977). The CES-D is a 20-item inventory assessing depressive symptoms. The CES-D scale scores have acceptable levels of internal consistency and convergent validity. The Portuguese version (Gonçalves & Fagulha, 2004) has coefficient alpha values that were between .87 and .92 in various samples. In the present sample, coefficient alpha was .91.

The Interpersonal Needs Questionnaire (INQ-15; Van Orden, Cukrowicz, Witte, & Joiner, 2012). The INQ-15 is a 15-item self-report measure that assesses two of the primary components of Joiner's (2005) interpersonal-psychological theory of suicide: perceived burdensomeness and thwarted belongingness. Items are answered on a 7-point scale ranging from 1 (*not at all true for me*) to 7 (*very true for me*). Van Orden et al. (2012) have reported results supporting the validity of scores from both scales. The Portuguese version of the questionnaire correlates significantly with both the Beck Hopelessness Scale and the Suicide Behaviors Questionnaire-Revised (Beck et al., 1974). Coefficient alpha reliabilities of .88 for the Perceived Burdensomeness scale scores and .80 for the Thwarted Belongingness scale scores are reported (Campos & Holden, in press). In the present study, Cronbach's alpha coefficient was .93 for the Perceived Burdensomeness scale and .80 for the Thwarted Belongingness scale.

The Psychache Scale (Holden et al., 2001). The Psychache Scale is a 13-item self-report measure developed to assess Shneidman's (1979) construct of psychological pain (psychache) as a relevant dimension for suicide risk. The Psychache Scale has adequate psychometric properties, with scale score alpha coefficients above .90 (e.g., Holden et al., 2001; Troister & Holden, 2012, 2013). The Portuguese version of the scale (Campos & Holden, in press) correlates with a measure of suicide risk and has a coefficient alpha reliability of .94. In the present sample, the alpha coefficient was .97.

Suicidal Behavior Questionnaire-Revised (SBQ-R; Osman et al., 2001). The SBQ-R was used to assess suicide risk based on the lifetime presence of suicide ideation and behaviors and comprises four multiple choice items: "Have you ever thought about or attempted to kill yourself?"; "How often have you thought about killing yourself in the past year?"; "Have you ever told someone that you were going to commit suicide, or that you might do it?"; and "How likely is that you will attempt suicide someday?". Using the SBQ-R, we evaluated four important clinical indicators of suicide risk: ideation/suicide attempts, recent ideation, communication of intention, and future probability. In its original version, the SBQ-R's scale scores' alpha reliability was .76. The Portuguese version (Campos et al., 2013; Campos & Holden, in press) has a coefficient alpha reliability of .74. In the present study, 24.6% of our sample had a score equal to or above the cutoff point of 7, as proposed by Osman et al. (2001), 28.1% reported some lifetime suicide ideation, and 7.4% indicated a previous suicide attempt. In the current study, Cronbach alpha reliability was .83.

Data Analytic Strategy

Structural equation modeling was used to test our models of suicide risk. All analyses were conducted using AMOS (version 21). In testing a model's fit and to compare models, we used the overall χ^2 test of exact fit and the following fit indexes: (a) the χ^2/df ratio, (b) the goodness-of-fit index (GFI), (c) the comparative fit index (CFI), (d) the normed fit index (NFI), (e) the

root mean square error of approximation (RMSEA), (f) the standardized square root mean residual (SRMR), and (g) the Akaike's information criterion (AIC).

We first examined multicollinearity among the variables, and results indicated the absence of excessive multicollinearity. The Kolmogorov-Smirnov z test examined distributions of variables and indicated nonnormality for some of them. Bootstrapping (with 5,000 samples to construct 95% bias-corrected confidence intervals) was therefore used to evaluate significance levels (e.g., Yung & Bentler, 1996). Bootstrapping also tested the significance of the indirect effects (e.g., Mallinckrodt, Abraham, Wei, & Russell, 2006).

Results and Discussion

Zero-order correlations between variables are displayed in Table 1. From the analyses that are summarized in Table 2, Model 1, our primary model, fit the data well ($\chi^2 = 54.243$, $\chi^2/\text{degrees of freedom} [df] = 1.937$, GFI = 0.950, CFI = 0.975, NFI = 0.950; RMSEA = 0.068, SRMR = 0.038, AIC = 108.243). This model explained 66% of the variance in suicide risk and provided a better fit relative to the alternative models.

Table 1
Zero-Order Correlations and Descriptive Statistics for Study Variables

Variable	1	2	3	4	5	6	7	8	9	<i>M</i>	<i>SD</i>
<i>Rejection</i>											
1. Mother	—									11.22	3.72
2. Father	.69**	—								13.63	4.57
<i>Depression</i>											
3. CES-D	.31**	.34**	—							17.36	10.67
<i>Interpersonal needs</i>											
4. Perceived burdensomeness	.38**	.34**	.51**	—						1.52	.91
5. Thwarted belongingness	.20*	.20*	.48**	.60**	—					2.69	1.03
<i>Psychache</i>											
6. Psychache scale	.26**	.29**	.76**	.53**	.51**	—				24.54	11.60
<i>Suicide risk</i>											
7. Ideation and attempt	.31**	.38**	.40**	.42**	.33**	.49**	—			1.64	.90
8. Recent ideation	.41**	.34**	.57**	.60**	.44**	.62**	.56**	—		1.35	.78
9. Communication	.27**	.27**	.45**	.41**	.29**	.51**	.64**	.66**	—	1.33	.62
10. Future probability	.25**	.26**	.50**	.47**	.41**	.58**	.55**	.64**	.55**	1.03	1.25

Note. *M* = mean; *SD* = standard deviation; CES-D = The Center for the Epidemiological Studies of Depression Scale.

* $p < .01$. ** $p < .001$.

Table 2
Results for the Primary and Alternative Models

Model	χ^2	χ^2/df	GFI	CFI	NFI	RMSEA	SRMR	AIC	Variance in suicide risk
1 primary model	54.243	1.937	0.950	0.975	0.950	0.068	0.038	108.243	66%
2	92.455	3.082	0.921	0.940	0.914	0.102	0.093	142.455	51%
3	71.240	2.375	0.936	0.960	0.934	0.082	0.050	121.240	64%
4	64.343	2.219	0.943	0.966	0.940	0.078	0.048	116.343	62%

Note. *df* = degrees of freedom; GFI = goodness-of-fit index; CFI = comparative fit index; NFI = normed fit index; RMSEA = root mean square error of approximation; SRMR = standardized square root mean residual; AIC = Akaike's information.

For the direct effects in Model 1, only the rejection to psychache link was not significant. For the indirect effects in Model 1, rejection related to suicide risk, via depression, and interpersonal needs. Rejection also significantly related to psychache and interpersonal problems. In addition, depression also related to psychache.

The aim of the present study was to evaluate, using structural equation modeling, a model of suicide risk and three alternative, competing models (given the exploratory nature of the study) in a nonclinical sample of adults, based on psychological variables deemed important in the literature. Results demonstrated that our proposed model provided a better fit than the competing alternative models. Burnham and Anderson (1998) suggest that if the AIC value for one model is 10 or more units lower than the AIC value for a second model, then there is strong evidence that the first model has better fit than the second model (see also Byrne, 2001). In our results, however, the AIC for Model 1 did not differ by more than 10 points compared with Model 4. However, based on other fit indexes, Model 1 did outperform Model 4. This advantage should be considered cautiously because indexes like the CFI or the NFI have limitations such as being affected by sample size. Given that one model is nested within another, the difference in chi-square values can test for relative fit (Kelloway, 1998), and in comparing Model 1 with Model 4, the difference in fit was significant, $\Delta\chi^2(1) = 10.10, p < .01$.

Note that one of the paths in Model 1 is not significant—the path between rejection and psychache. As an ancillary analysis, we tested a similar model removing this path and having perceived burdensomeness and thwarted belongingness as independent indicators (given some similarity between Models 1 and 4). Fit indexes were better but not substantially ($\chi^2 = 46.300$, $\chi^2/df = 1.715$, GFI = 0.957, CFI = 0.981, NFI = 0.957; RMSEA = 0.059, SRMR = 0.038, AIC = 102.300). The percentage of explained variance was smaller, 64%, and differences in AIC were not notable, but the $\Delta\chi^2$ value was significant.

From our results and consistent with the literature, rejection related directly to suicide risk and related indirectly via depression (e.g., Campos et al., 2013; Campos & Holden, in press) and interpersonal needs. Contrary to expectations, our findings indicated that rejection did not directly relate to psychache, only indirectly via depression and interpersonal needs. This indicates that individuals who perceive that their parents were rejecting may be vulnerable to suicide, experiencing depression, and feelings of burdensomeness and a lack of belongingness that subsequently produce psychological pain. Thus, it is not the feeling of being rejected that leads people to experiencing despair and suffering, but it is their potential to feel depressed, isolated, and like being a burden to others that makes them vulnerable to an inner state of psychological anguish and pain. Of note, based on current results, people who feel rejected by their parents may be at risk for suicide, even when they do not feel depressed, isolated, or like being a burden to others; therefore, we suggest that this dimension should be more fully researched in future studies.

According to our results and consistent with previous research (e.g., Wong et al., 2011), it may be the perceived burdensomeness component of interpersonal needs that augments the risk for suicide. Current results also align with previous findings (e.g., Campos & Holden, in press) indicating that psychache and interpersonal needs mediate the relationship between depression and suicide risk. Not all depressed individuals present with suicide ideation and behaviors; rather, it is probably the potential of depression to induce psychological pain, isolation, and feelings of being a burden that eventually leads an individual to become suicidal. From our findings, it is important to assess psychache and interpersonal needs, particularly perceived burdensomeness, when clients present with depressive symptoms because it may be these former variables that lead to the client being vulnerable to suicidality.

Limitations

The present study has potential limitations. First, our data were drawn from a relatively small convenience sample and may not be fully representative of the Portuguese population. Second, our assessments related to suicide risk and may show some restriction of range relative to clinical samples. The use of a nonclinical sample may be an issue when focusing on proximal risk factors for suicide; however, our purpose was to study the contribution of important psychological

variables in a general sample before investigating a high-risk population. Studying clinical phenomena in nonclinical samples is an important precursor for identifying possible risk factors, even when individuals present low risk. Third, we relied on self-report measures to collect data, including information regarding suicide risk.

Fourth, because our research design was cross-sectional rather than longitudinal, limited causal inferences can be drawn, particularly regarding the distal variable of perceived rejection and depression and suicide risk. Experiencing vulnerability and distress might confound and influence the recollection of participants' childhood experiences. However, the literature indicates that although distressed mood may influence retrospective reporting, such reports are generally quite accurate (e.g., Blatt, 2004). Fifth, our choice of variables to investigate may have omitted other important factors for suicide risk, for example, external variables such as sociodemographics, negative life events, and important clinical variables such as the presence of psychiatric disorders.

Finally, the EMBU has limitations for assessing perceived parental rejection because it assumes a two-parent (heterosexual) household. Future research with measures that are not self-report (e.g., interviews), alternative sampling methods (e.g., random stratified), clinical samples, longitudinal designs, and other variables will serve to delimit the generalizability of our present findings.

Conclusion

In summary, our tested models accounted for a large proportion of variance in suicide risk in a nonclinical sample. Our results indicate the importance of assessing perceptions of parental rejection, psychache, and interpersonal needs and evaluating depressive symptoms when considering potential risk for suicide in nonclinical individuals.

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